01 / PROJECT CHARTER

A management system built around daily operations

PasadaTrack

Jeepney Boundary Collection, Driver Scheduling, Attendance and GPS Tracking System

PREPARED FOR	REVISION	REVIEW DATE
Jeepney operator / project sponsor	1.1 - proposed revised baseline	10 October 2026

Purpose and expected outcome

Provide a mobile-friendly web application that records planned and actual driving, boundary obligations, collections and attendance, with a fleet map connected to real GPS devices. Keep location, attendance, driving confirmation and payments as separate evidence and workflows.

What this revision changes

The original 9 October 2026 proposal is aligned with the repository reviewed on 10 October. This revision includes driver management and all original requirements, adds live GPS, corrects the technical stack and data terminology, and separates implemented screens from unfinished operational workflows.

Planning assumptions

The pilot still assumes one operator, one operating location, one developer and staff available for weekly reviews. Existing team infrastructure does not establish readiness for multiple independent operators. The operator supplies master records, business rules and approved opening balances. Biometric hardware is unselected; Teltonika FMC920 is a GPS reference, not a confirmed procurement choice.

Success measures

During a five-operating-day pilot, confirmed driving produces exactly one obligation; receipts reconcile with the signed collection log; schedules have no unresolved conflicts; scans are processed or flagged; real mapped GPS devices update the fleet display; and staff independently complete the daily workflow.

> **Baseline status**
>
> A proposed revised baseline, not a delivery or procurement commitment. Sponsor agreement, device feasibility, production configuration and a revised estimate remain required. The original eight-week / 195-240-hour estimate did not include live GPS.

02 / SCOPE AND REQUIREMENTS

The release backlog

Retain FR-01 to FR-09 and add FR-10 for live GPS. Existing foundations reduce setup work; they do not satisfy the acceptance criteria by themselves.

ID / priority	Requirement	Acceptance evidence
FR-01 / Must	Manage drivers, jeepneys, routes and default boundary rates.	Create, edit and deactivate records; retain historical references.
FR-02 / Must	Schedule drivers and jeepneys by date and shift.	Reject active driver/vehicle overlaps and blocked vehicle periods.
FR-03 / Must	Confirm actual driving, substitutes and unscheduled work.	Retain planned details; identify the actual driver and confirmation actor.
FR-04 / Must	Assess boundaries and approve signed adjustments.	Freeze the historical base rate; record approver, reason and signed amount.
FR-05 / Must	Record payments, allocations and reversals.	Support partial and multi-day settlement without excess or duplicate allocation.
FR-06 / Must	Show collections, balances and Excel exports.	Distinguish driving dates from collection dates; exports reconcile.
FR-07 / Must	Enforce operational permissions and audit history.	Deny unauthorized actions; trace approvals, corrections and reversals.
FR-08 / Integrated pilot	Map biometric device users and ingest events.	Authenticate imports; deduplicate events; flag unknown identities.
FR-09 / Integrated pilot	Process attendance against schedules with manual fallback.	Handle overnight shifts, missing scans and corrections with evidence.
FR-10 / Integrated pilot	Connect real GPS devices to the fleet map.	Store authenticated vehicle positions; refresh display; show sample time and stale/offline status.

Outside this release

Passenger fares, payroll, automatic bank reconciliation, SMS reminders, driver self-service, multiple independent operator rollout, maintenance/expense modules and full profit reporting remain deferred. Vehicle unavailability is included for scheduling; a full maintenance system is not. Boundary collections alone are not profit.

> **Quality requirements**
>
> Responsive phone layouts, decimal money handling, transactional allocation, UTC timestamps with Asia/Manila business dates, team isolation, authenticated integrations, retry/deduplication, separate backups and a restore drill. Proposed recovery targets remain 4 hours to restore and at most 24 hours of database loss, subject to sponsor approval.

02 / CURRENT IMPLEMENTATION STATUS

What exists, and what is still missing

Status is based on source inspection and the connected database schema, not a completed acceptance run. Review date: 10 October 2026.

Existing = reachable implementation found **Foundation** = models/migrations only
Missing = required workflow not found **Unverified** = operational evidence not available

Area	Current evidence	Remaining work
Accounts / teams	**Existing:** auth, settings, teams, invitations and membership controls.	Domain permissions, approvals and operational roles.
Tracking	**Existing demo:** map, search, filters, sample vehicle/driver details.	Real devices, ingestion, stored GPS fixes and automatic refresh.
FR-01 masters	**Foundation:** driver, jeepney and route models, factories and migrations.	Master-record screens, validation, deactivation and access control.
FR-02 scheduling	**Foundation:** assignments, statuses and vehicle unavailability.	Calendar, overlap checks, availability checks and cancellation.
FR-03 actual driving	**Foundation:** confirmed driving record and rate snapshot.	Atomic confirmation, substitutes and unscheduled-driving workflow.
FR-04 boundaries	**Foundation:** obligations and approved-adjustment storage.	Assessment, approval, balance computation and traceable entry.
FR-05 payments	**Foundation:** payments, allocations, reversals and history guards.	Collection screens, transactional limits, concurrency and reversal processing.
FR-06 reports	**Missing:** no collection/balance reports or Excel workflow found.	Date-based totals, reconciliation, statements and matching exports.
FR-07 audit	**Foundation:** audit-entry storage; team permissions exist.	Automatic domain audit capture, viewer and permission enforcement.
FR-08 / FR-09	**Foundation:** devices, users, raw events, sessions and corrections.	Manual attendance, import/connector, processing, matching and exception review.
Operations	**Unverified:** no backup/restore workflow found in the inspected project.	Deployment readiness, monitoring, restore proof, training and pilot.

Important limits of the evidence

The connected SQLite database contains account/team tables; the 17 new domain migrations are present in the working tree but are not applied there. Tracking returns synthetic samples locally/testing and an unconnected state elsewhere. New foundation tests use Pest syntax while installed tooling is PHPUnit; test success has not been verified. Existing uncommitted work is included in this review, not represented as an accepted release.

03 / OPERATIONS AND DATA DESIGN

Separate plans, facts and money

Daily operating procedure

1. Plan assignments and blocked vehicle periods.
2. Capture attendance by scan or authorized manual entry.
3. Confirm actual driving, substitutes or cancellation.
4. Create the confirmed driving record and boundary obligation atomically.
5. Receive money and allocate it to eligible obligations.
6. Review return attendance, missing scans and GPS/device exceptions.
7. Reconcile receipts, balances and outstanding exceptions.

Data group	Repository terminology / responsibility
Accounts / assets	users, teams, team_members, drivers, jeepneys, routes; route default boundary amounts.
Scheduling	assignments, vehicle_unavailabilities; planned driver, vehicle, route, UTC shift window and local operational date.
Driving / dues	driving_records, boundary_obligations, boundary_adjustments; actual driver, immutable rate snapshot and approved changes.
Collections	payments, payment_allocations, payment_reversals; receipts, settlement links and traceable reversal facts.
Attendance	biometric_devices, biometric_device_users, biometric_events, attendance_sessions, attendance_corrections.
Traceability / GPS	audit_entries exists as a foundation. GPS device mappings and stored location history remain to be designed after interface feasibility.

Financial and ownership rules

One confirmed assignment has at most one driving record and one obligation. Payments settle obligations of the same actual driver and team. Amount due = saved base amount + approved signed adjustments. Outstanding = amount due - allocations from unreversed payments. Services must validate decimal amounts and allocation limits transactionally; schema relationships alone do not implement these rules.

Design discrepancies to resolve

The current driving record requires an assignment, while the original plan permits unscheduled work: define the ad hoc assignment path before confirmation implementation. Attendance currently stores in/out event references without a schedule/driving-record link; matching and evidence grouping remain pending. Enrollment/revocation history and explicit ingestion processing/exception status are also unfinished.

> **Historical integrity**
>
> Deactivate master records and preserve referenced financial history. Keep raw scan events unchanged. An append-only model guard is a foundation, not proof that every write path and database permission preserves history.

04 / BIOMETRIC INTEGRATION

Prove the device interface before committing

Terminal → supported API/export or local connector → authenticated ingestion → attendance processing → operator review.

Stage	Work and exit condition
Compatibility gate	Confirm a real supported interface, stable event identifiers, device clock/timezone, verification method and offline history. Demonstrate one real event reaching a test environment before procurement acceptance.
Enrollment	Enroll on the terminal and map device user IDs to same-team drivers. Enforce device-scoped identity uniqueness and define enrollment/revocation history.
Event ingestion	Authenticate the source; preserve occurrence and receipt times plus original metadata; deduplicate retries and flag unknown users or invalid events.
Attendance processing	Match applicable shift windows. Review overnight shifts, repeated scans, unscheduled scans and missing check-outs without fabricating attendance.
Recovery / fallback	Capture authorized manual attendance with actor and reason. Retry unsynced events where supported; reconcile late scans with manual records without duplicating sessions.
Privacy / access	Keep templates/matching on the device where supported. Store identifiers and attendance results in the app. Assign enrollment access and obtain sponsor decisions on retention, deletion and driver notice.

Current baseline

Device/user/event/session/correction migrations and models are authored. There is no usable enrollment, import, attendance-processing or exception-review workflow. Database event uniqueness is useful groundwork; authenticated import, retry behavior and matching still require implementation and acceptance evidence.

Rules protecting daily records

A scan confirms attendance only. Operator confirmation identifies who actually drove a jeepney. Collection staff separately record money received. Correcting attendance must retain the original event evidence and correction history.

> **Device-dependent release decision**
>
> If compatibility fails or hardware is delayed, a core pilot may use authorized manual attendance only after sponsor acceptance of the exclusion. The biometric milestone remains pending; the revised integrated release is not represented as complete.

04 / GPS INTEGRATION - FR-10

Turn the existing map into real tracking

Reuse the fleet map. Replace the synthetic snapshot with verified, team-scoped locations from real devices.

Stage	Work and exit condition
Hardware / interface gate	Confirm the actual tracker model, payload/interface, credentials, connectivity, timestamps and offline behavior. Teltonika FMC920 is only the current code reference. Capture a real fix in the test environment.
Device mapping	Register an authorized GPS source and map it to a jeepney in the correct team. Reject or flag unrecognized identities; do not guess vehicle ownership.
Ingestion / storage	Authenticate the source, validate locations and timestamps, retain fix time and receipt time, and handle duplicate or delayed messages deterministically. Keep location history separate from financial facts.
Fleet display	Feed real latest locations into the existing map/search/details flow. Refresh automatically, show the fix timestamp and freshness, and retain useful last-known locations with honest stale/offline labeling.
Monitoring / recovery	Expose ingestion failures, unrecognized devices and loss of recent updates. Demonstrate reconnect/offline recovery supported by the selected interface. Keep demo mode clearly labeled and restricted.

Acceptance boundary

A vehicle movement from a mapped real tracker appears on the map after a new fix is received. The display distinguishes fresh, stale and absent locations. Duplicate or older fixes do not corrupt latest-location selection. Unauthorized sources and cross-team access are rejected. Device outages and reconnection are observable.

Keep evidence separate

GPS location does not establish driver identity, confirm a driving assignment, record attendance or create a payment. Actual-driver details come from the reviewed assignment/confirmation workflow, not assumptions inferred from location.

Scope and dependency impact

Live GPS moves from the original out-of-scope list into this release. Tracker hardware, connectivity/SIM or platform fees, installation, integration and support costs need separate estimates. Trip playback, geofencing, route-deviation alerts and advanced telematics are not added by this requirement.

> **GPS release gate**
>
> Approve hardware/interface feasibility and a revised estimate before committing to delivery dates. If GPS is excluded from a core pilot, retain FR-10 as pending and record the sponsor-approved exception explicitly.

05 / WORK BREAKDOWN AND MILESTONES

Plan from the remaining work

Use dependency-based milestones rather than treating the original eight-week proposal as an unchanged commitment. A single developer implements sequential work; device access and sponsor reviews can create calendar waits.

Milestone	Remaining work	Exit deliverable
M0 / Rules and feasibility	Confirm rates, shifts, payment allocation rules, opening balances, roles, biometric interface and GPS interface. Review existing foundations and test-runner mismatch.	Approved rules, device proof and revised effort/budget baseline.
M1 / Masters and permissions	Complete driver/jeepney/route workflows and operational authorization; validate and apply domain migrations in a controlled test environment.	Staff manage active records and unauthorized actions fail.
M2 / Scheduling and driving	Calendar, overlap/availability checks, cancellation, substitutes and unscheduled-driving path; atomic confirmation.	Reviewed schedules and one preserved actual-driving fact per confirmation.
M3 / Financial workflow	Boundary assessment, approved adjustments, collection/receipts, partial and multi-day allocations, reversals and audit capture.	Collector reconciles receipts and balances; concurrency/integrity scenarios pass.
M4 / Reports and core readiness	Collections/balances, Excel exports, manual attendance with reasons, responsive flows, backups/restore and staff training.	Core acceptance; a core-only pilot requires explicit integration exclusions.
M5 / GPS integration	Real device mapping, ingestion/storage, refreshed map, freshness and failure monitoring.	FR-10 proven with real fixes, outage and recovery scenarios.
M6 / Biometrics	Enrollment mapping, authenticated ingestion, deduplication, attendance matching and exceptions.	Real scan events and manual/late-event reconciliation demonstrated.
M7 / Integrated pilot	Five operating days, daily reconciliation, device/recovery checks, defect closure and handover.	Signed integrated acceptance, exclusions if any, and support log.

Estimate control

Original baseline: 8 working weeks and 195-240 developer hours, assuming 25-30 hours per week. Retain these as historical figures only. The revised remaining effort must credit reusable work, include live GPS, resolve test readiness, and include device integration/acceptance. No new fixed date or total is established by this document.

> **Dependency gates**
>
> Reports depend on correct financial allocations. Device integration depends on real interface access. Integrated acceptance depends on reconciled live pilot records and verified GPS/biometric evidence. Record each approved estimate or scope change as a new baseline revision.

06 / RESPONSIBILITIES AND GOVERNANCE

Name owners and accept evidence

Activity	Accountable	Responsible / consulted
Scope, rules and revised baseline	Operator / sponsor	Developer drafts; collector and driver representative review.
Technical quality and integration	Developer / project lead	Developer implements/tests; sponsor resolves business decisions.
Payment reconciliation	Operator / sponsor	Collector compares app totals with receipts and signed cash log.
Biometric / GPS feasibility	Operator / sponsor	Developer verifies; vendor/device contact supplies interface access and installation support.
Acceptance / release decision	Operator / sponsor	Collector/operator run scenarios; developer resolves defects.
Daily operations / backups	Operator / sponsor	Named staff review exceptions/backups; developer provides handover and incident support.

Communication and change control

Hold a weekly 30-minute demonstration and review: completed evidence, unresolved decisions, risk updates and next milestone. During the pilot, reconcile daily. Maintain a backlog, decision log, risk register and defect list. Each material change records acceptance criteria, effort, dependencies and scope/schedule/budget impact; the sponsor approves its revised baseline.

Definition of done

A feature has a reachable workflow, correct business behavior, meaningful passing regression tests, checked authorization/team isolation, required audit evidence, updated operator instructions and operator review. Database tables, comments describing intended services and screen mockups alone do not establish completion.

Actual development structure

The reviewed project uses PHP 8.4, Laravel 13.35.0, React 19, Inertia 3.5.1, TypeScript, Tailwind 4 and Laravel Wayfinder. The connected database is SQLite. Filament and Livewire are not the implemented frontend; the original MySQL statement is not the current database baseline.

Follow existing controllers, requests, policies, actions and grouped models. Add transactional domain workflows and isolated device adapters as needed; do not mandate new repositories or a second UI framework. Queues and scheduled operational work must be configured and verified. Production database/hosting selection remains a sponsor/technical decision.

> **Verification readiness**
>
> Installed test tooling is PHPUnit 12.5.38, while the new foundation file contains Pest-style calls. Resolve this compatibility issue without assuming those tests pass. Align CI with supported package/PHP versions before using it as release evidence. No tests were executed for this document-only review.

07 / RESOURCES AND BUDGET

Separate historical allowances from new costs

These are the original illustrative allowances, not current quotations. This revision authorizes no spending and provides no vendor price claim.

Item	Original allowance / revised treatment
Application hosting	PHP 500/month; confirm production database, queue workers, scheduled tasks and integration load.
Domain	PHP 1,000/year; existing domain/subdomain may be reused.
Separate backups	PHP 200/month; verify retention, monitoring and restore capability.
Biometric terminal	PHP 8,000 once; actual compatible model/interface requires a quote and feasibility proof.
Local connector equipment	PHP 3,000 once; only if the selected interface/network requires a dedicated host.
Power / network accessories	PHP 1,500 once; confirm site and installation needs.
Developer labor	Original 195-240 hours x agreed rate; revised remaining effort including GPS is pending.
Vendor API / SDK / support	Quote required; confirm access, licensing, support and installation terms.
GPS trackers / installation	New scope: quantity, model, installation and initial connectivity quote required.
GPS recurring service	New scope: SIM/data or platform fees and support quote required.
GPS integration / verification	New scope: estimate adapter, storage, refreshed map, monitoring and real-device acceptance effort.

Historical first-year illustration

Hosting (500 x 12) + backups (200 x 12) + domain (1,000) + terminal (8,000) + connector (3,000) + accessories (1,500) = PHP 21,900. Original 15% contingency = PHP 3,285; total = PHP 25,185.

This is not the revised release budget. It excludes labor, vendor licensing/support and all newly included GPS costs. Reused equipment or changed infrastructure also changes the illustration. Obtain actual quotations, then calculate contingency on the approved revised cost basis.

People and equipment

Provide one developer, an operator decision-maker, a collector, a driver representative and device/vendor contacts. Supply a development laptop, test phone, sample records, terminal/tracker access, network connectivity, suitable hosting and a separate backup destination. Name enrollment, tracker installation and ongoing support owners.

08 / RISKS AND SWOT

Protect balances and device-dependent delivery

Risk / priority	Response and owner
Incomplete foundation / High	Developer validates migration readiness and runnable tests; do not count models as delivered workflows.
Incorrect balances / High	Developer enforces decimal validation and transactional limits; collector reconciles receipts and tests reversals/concurrency.
Unsupported devices / High	Developer proves biometric and GPS interfaces; sponsor changes hardware or approves a documented core-only exception.
Device / network outage / High	Named operator reviews missing data; use traceable manual attendance and supported retry/reconnection behavior.
Unauthorized / cross-team access / High	Developer enforces domain permissions and device authentication; sponsor restricts staff and integration access.
Data loss / High	Automate separate backups; named operator checks failures; developer demonstrates usable restoration.
Misleading tracking status / High	Show real fix times and freshness; never pass synthetic samples off as live or infer driving/payment from GPS.
Scope and capacity / Medium	Project lead estimates remaining work including GPS; sponsor approves costs and milestones before commitments.
Legacy records / adoption / Medium	Collector validates opening balances and retains provenance; operator pilots realistic phone workflows and daily reconciliation.

SWOT and strategic response

Area	Assessment and action
Strengths	Existing account/team infrastructure, reusable fleet-map UI and authored domain foundations. Reuse them while prioritizing trustworthy collections.
Weaknesses	Single-developer capacity, incomplete workflows and unverified test readiness. Finish one end-to-end operational path at a time.
Opportunities	Connect fleet visibility with reviewed schedules and attendance; validate maintenance, expenses and driver access after pilot acceptance.
Threats	Device incompatibility, poor connectivity, inconsistent records and new GPS costs. Prove interfaces early and keep explicit fallback decisions.

Risk review

Review weekly, with a named owner, due date and mitigation evidence. Escalate threats to reconciliation, access, device feasibility or milestone dates. The operator approves device-data retention and notice procedures before rollout.

09 / TESTING AND ACCEPTANCE

Verify workflows, not just structures

Scenario	Required result
Masters / permissions	Create/edit/deactivate without losing history; unauthorized and cross-team domain actions fail.
Scheduling / substitutes	Reject driver/vehicle overlaps and unavailable periods; permit non-overlap. Cancellation creates no dues; actual substitute remains traceable.
Confirmation / rates	Atomic, repeat-safe confirmation produces one driving fact/obligation; later default-rate changes leave historical amounts fixed.
Payments / allocation	PHP 700 against PHP 1,000 leaves PHP 300. Multi-day settlement obeys same-driver/team ownership and both payment/due limits.
Concurrency / reversal	Concurrent requests cannot overpay or double-spend. Reversal reopens balances and records actor/reason without deleting receipts.
Reports / Excel	Driving-date dues and collection-date receipts reconcile separately; exported rows/totals match displayed filters and balances.
Biometric identity / retries	Reimport deduplicates; unknown users go to review and create no incorrect attendance; unauthorized imports fail.
Attendance / recovery	Overnight matching, missing check-outs and unscheduled scans are reviewable; manual/late-scan recovery avoids duplicate attendance.
Real GPS / timestamps	A mapped real-device movement appears after a received fix; display shows fix time, fresh/stale/no-location state and useful last-known data.
GPS failures / isolation	Duplicates/out-of-order messages do not regress latest location. Unknown/unauthorized devices are rejected or flagged; outages/reconnect and team isolation are verified.
Backup / restore	Restore usable records within approved recovery targets; preserve and reconcile transactions entered after a release rollback.

Release acceptance

Run meaningful PHPUnit coverage for implemented behavior and boundary failures, then operator scenarios with approved sample records and a five-day live pilot. Resolve defects that corrupt balances, bypass permissions or block the main workflow. Minor defects need an owner and due date. Device-dependent exclusions require sponsor acceptance and remain pending requirements.

> **Acceptance record**
>
> Record release/build, test period, scenario outcomes, reconciled totals, real device/interface evidence, restore results, remaining defects, approved exclusions, sponsor name and acceptance date. The current static review is not a test or release acceptance certificate.

10 / DEPLOYMENT AND HANDOVER

Release with a reconciled operating record

Before launch

Confirm production configuration, HTTPS, domain permissions, team isolation, UTC/Asia-Manila date handling, migration readiness, workers/scheduled jobs, integration credentials, device mappings, monitoring and separate backups. Prove restoration. Import operator-approved opening balances with provenance rather than fabricated payments. Train staff using realistic shifts, substitutes, collections and device failures.

Pilot, fallback and rollback

Run five operating days while retaining paper/receipt logs as the reconciliation reference. Resolve discrepancies daily. If a critical defect appears, suspend the affected workflow and retain a fallback log. Roll back the application only with compatible migrations; never overwrite new payments with an older database without preserving/reconciling them. Import fallback entries with source references after recovery.

Handover deliverable	Required contents
Application / data design	Repository/release reference, actual schema and constraints, architecture and deployment instructions.
Operator guide	Masters, scheduling, substitutes, confirmation, dues, partial payments, reversals, daily closing and Excel export.
Device guides	Verified biometric/GPS models and interfaces; identity/vehicle mappings, clock checks, sync, stale/offline review and fallback.
Operations / support	Backup/restore evidence, monitoring, incident contacts, secure access transfer, pilot acceptance, exclusions and support terms.

Decisions required to finalize the baseline

Confirm rates and exceptions; shifts/rest days; overdue allocation policy; unscheduled/substitute driving process; operational roles; biometric model/interface; actual GPS tracker/interface and quantity; opening balances; retention; development hours/start date; production database/hosting; revised cost/effort; and support ownership.

Review provenance and revision record

Source: original PasadaTrack Project Management Plan v1.0, 9 October 2026 (10 pages). Reviewed repository: routes, controllers, React pages, domain models/migrations, package manifests, tests and CI. Laravel Boost confirmed framework/package information and connected SQLite schema/migration history. All findings refer to the 10 October working-tree snapshot; test and live-device success were not asserted.

Key evidence: `app/Http/Controllers/DashboardController.php` and `app/Data/Tracking/DemoFleet.php` show synthetic/unconnected tracking. Domain migrations/models show foundations; routes/pages show missing management workflows. `tests/Feature/Foundation/SchemaTest.php` uses Pest-style calls; the lockfile declares PHPUnit, not Pest. The original PDF and existing source changes remain preserved.

> **v1.1 revision**
>
> Adds current implementation status and live GPS; corrects the stack/schema baseline; replaces unchanged calendar promises with gated milestones; updates budget, risks and acceptance evidence. Sponsor approval fields remain unconfirmed.